Software Test Plan (STP) - Blood Bank Management System

Project: Blood Bank Management System
Version: 1.0
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1. Introduction

Purpose: This document defines the test plan for the Blood Bank Management System (BBMS) v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing.

Scope: Testing covers user registration and authentication, donor profile and eligibility, donation records, blood/component inventory, availability search, blood requests, approval, reservation, issue/dispatch, alerts, reports, audit logs, and administrative functions. Laboratory testing procedures and hospital internal systems are excluded.

References: BBMS SRS v1.0, Design Specifications v1.0, project requirements, and applicable security requirements.

Definitions: BBMS (Blood Bank Management System), UI (User Interface), API (Application Programming Interface), RBAC (Role-Based Access Control), HTTPS (Hypertext Transfer Protocol Secure), TLS (Transport Layer Security), RTM (Requirements Traceability Matrix).

2. Test Items

- User registration and authentication module
- Donor profile and eligibility module
- Donation appointment and donation record module
- Blood/component inventory module
- Blood availability search module
- Blood request, approval, reservation, and issue/dispatch modules
- Alert, reporting, audit, and administrative interface

3. Features to be Tested

Features mapped to SRS requirement IDs:
- BBMS-F-001: User registration
- BBMS-F-002: User authentication

- BBMS-F-003: Role-based permissions
- BBMS-F-004: Profile management
- BBMS-F-005: Donor eligibility and slot booking
- BBMS-F-006: Donation record management
- BBMS-F-007: Inventory update after donation
- BBMS-F-008: Blood availability search
- BBMS-F-009: Blood request submission
- BBMS-F-010: Request approval/rejection
- BBMS-F-011: Compatible unit reservation
- BBMS-F-012: Issue/dispatch and inventory update
- BBMS-F-013: Prevent expired/unavailable/incompatible issue
- BBMS-F-014: Low-stock and expiry alerts
- BBMS-F-015: Reports, audit logs, and admin management
- BBMS-NF-001: 95% of normal requests complete within 3 seconds
- BBMS-NF-002: At least 99.5% monthly availability
- BBMS-NF-003: Transaction and inventory data integrity
- BBMS-NF-004: At least 200 concurrent active users
- BBMS-NF-005: Usability, validation, accessibility, and responsive UI
- BBMS-SR-001 to BBMS-SR-005: Secure transport, password hashing, RBAC, session security, and audit logging

4. Features Not to be Tested

- Laboratory blood testing procedures and medical screening outside the software system
- Internal hospital systems beyond the defined BBMS interfaces
- Physical blood collection, storage, and transportation processes
- Underlying database engine internals and server operating-system internals

5. Test Approach / Strategy

Levels:
- Unit tests (module-level)
- Integration tests (Web UI/API ↔ database and connected application services)
- System tests (end-to-end BBMS workflows)
- Acceptance tests (UAT)

Types:
- Functional testing (registration, authentication, donation, inventory, requests, issue, alerts, reports)
- Regression testing
- Performance testing (response time and concurrent users)
- Usability testing (UI clarity, validation, accessibility, responsive behavior)
- Data integrity testing (inventory, request, reservation, and issue consistency)

Entry Criteria: Stable build delivered, representative test data available, test environment ready, and required database/API services available.

Exit Criteria: 100% of planned test cases executed, 0 critical defects open, all high-priority acceptance criteria satisfied, and traceability completed.

5.1 Security Validation

- Validate HTTPS/TLS for browser-to-server and API communication
- Verify passwords are securely hashed and never stored in plaintext
- Verify role-based access control for Donor, Requester, Blood Bank Staff, and Administrator roles
- Verify session timeout, logout, and secure session invalidation
- Verify audit logging of critical user, inventory, approval, issue, and administrative actions
- Perform input validation and security testing for authentication, requests, and administrative forms

6. Test Environment

Hardware: Development/test systems capable of running the web application and database server; client systems for desktop/mobile browser testing.
Software: BBMS v1.0, application services using Python (Django) and/or JavaScript (Node.js), relational database, and supported web browsers.
Tools: Selenium (UI automation), Postman (API testing), JMeter (performance/load testing), and Jira or equivalent defect tracking.
Test Data: Dummy donor/user profiles, blood groups, donation records, blood units, hospital/requester accounts, requests, reservations, issue transactions, and alert conditions.

7. Test Schedule

Milestones:
- Test case design: 28-Sep-2026
- Environment setup: 30-Sep-2026
- Test execution start: 31-Sep-2026
- Test execution end: 5-Oct-2026
- UAT: 06-Oct-2026 to 09-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Requirements Traceability Matrix (RTM)
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility

Test Engineer	Gurudath A N	Design/execute tests and log defects
QA Lead	M Sangamesh Prasad	Prepare plan and coordinate testing
Product Owner	R Sai Pranav Reddy	Review results and approve readiness
Developer	Abhiram Kolalapudi	Fix defects and support triage

10. Risks and Mitigation

Risk	Mitigation
Delay in stable BBMS build delivery	Request early smoke builds and freeze a test baseline before full execution
Test environment or database downtime	Maintain a backup test environment/database snapshot and restore procedure
Invalid or inconsistent test data	Prepare controlled dummy donor, unit, request, and transaction datasets

11. Assumptions & Dependencies

- BBMS application build and database will be available for testing before execution
- Representative donor, blood unit, requester, and transaction test data will be provided before execution
- Required application services, APIs, and database connections will remain available during testing
- Role and eligibility rules provided in the SRS will be treated as the test baseline

12. Suspension & Resumption Criteria

Suspend testing if:
- Test environment or core application services are unavailable for >4 hours
- Build is too unstable (blocks >30% of planned test cases)
- Critical data integrity or security defects make further execution unsafe

Resume testing if:
- Blocking defects are resolved or an approved stable build is deployed
- Environment and required services are stabilized
- Test data and database state are restored to a valid baseline

13. Test Case Management & Traceability

RTM ensures mapping of SRS requirements to test cases.
Examples:
- BBMS-F-001 (User registration) → TC-REG-01
- BBMS-F-002 (Authentication) → TC-AUTH-01
- BBMS-F-003 (RBAC) → TC-AUTH-02
- BBMS-F-007 (Inventory update) → TC-INV-01
- BBMS-F-009 (Blood request) → TC-REQ-01
- BBMS-F-012 (Issue/dispatch) → TC-ISS-01
- BBMS-NF-001 (Response time) → TC-PERF-01
- BBMS-NF-004 (Concurrent users) → TC-SCAL-01
- BBMS-SR-001 to BBMS-SR-005 → TC-SEC-01 to TC-SEC-05

14. Test Metrics & Reporting

Metrics collected:
- % test cases executed
- % passed/failed/blocked
- Defect density
- Defect aging
- Requirement coverage
- Security requirement coverage
- Performance response-time and concurrency results

Reports:
- Daily execution status
- Defect status report
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
Test Engineer	Gurudath A N	
Product Owner	R Sai Pranav Reddy	
Developer	Abhiram Kolalapudi	
QA Lead	M Sangamesh Prasad	